

1. Administrative & Study Identification

Full Study Title: A Study of Adjuvant Pembrolizumab/Vibostolimab (MK-7684A) Versus Pembrolizumab for Resected High-Risk Melanoma in Participants With High-Risk Stage II-IV Melanoma **Short Title/Acronym:** KEYVIBE-010 **Protocol Number:** MK-7684A-010 **NCT Number:** NCT05665595 **Posting ID:** 908417009514788791 **Sponsor/Company Name:** Merck Sharp & Dohme (MSD) / Merck **Trial Status:** COMPLETED **Investigational Product:** Coformulated Vibostolimab and Pembrolizumab (MK-7684A) vs. Pembrolizumab **Phase of Development:** Phase III **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To compare the efficacy of adjuvant pembrolizumab/vibostolimab to pembrolizumab alone with respect to recurrence-free survival (RFS) as assessed by the investigator in participants with high-risk resected Stage IIB, IIC, III, and IV melanoma. **Secondary Objectives:** To evaluate distant metastasis-free survival (DMFS) by investigator review, overall survival (OS), safety and tolerability, and quality of life (QOL).

2.2 Background & Rationale Pembrolizumab is an established standard-of-care adjuvant therapy for high-risk resected melanoma that significantly improves RFS. However, a high medical need remains to further prevent tumor recurrence. Vibostolimab is a novel anti-TIGIT monoclonal antibody designed to restore antitumor immune responses. This study was designed to test whether combining pembrolizumab with vibostolimab could provide superior RFS outcomes compared to pembrolizumab alone in the adjuvant setting.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Pembrolizumab alone) **Blinding:** Double-blind **Randomization:** 1:1, stratified by risk-based staging and geographical region

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 1,594 participants (1,402 randomized by the interim data cutoff) **Number of Sites:** 205 global sites (hospitals and cancer centers) **Estimated Study Start:** 19-Jan-2023 **Primary Completion Date:** 06-Mar-2024 (Trial discontinued for futility; patients unblinded and offered crossover to pembrolizumab monotherapy)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female patients ≥ 12 years of age (weighing ≥ 40 kg if < 18 years). 2. Surgically resected and histologically/pathologically confirmed stage IIB, IIC, III, or IV cutaneous melanoma per AJCC 8th edition, with no evidence of metastatic disease after resection. 3. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. 4. No more than 12 weeks elapsed between final surgical resection and randomization.

4.2 Exclusion Criteria 1. Diagnosis of ocular, mucosal, or conjunctival melanoma. 2. Prior systemic therapy for melanoma beyond complete resection. 3. Prior allogeneic tissue or solid organ transplant. 4. Active autoimmune disease, immunodeficiency, or receiving chronic systemic steroids (>10 mg prednisone or equivalent).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Investigational Drug: Pembrolizumab (Trade Name: Keytruda) **ATC Code:** L01FF02 **Dosage/Frequency:** Intravenous coformulated vibostolimab 200 mg with pembrolizumab 200 mg OR pembrolizumab 200 mg alone every 3 weeks (adolescent participants receive 2 mg/kg up to a maximum of 200 mg). **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Up to 17 cycles (approximately 1 year), or until disease recurrence, unacceptable toxicity, or withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for treatment-related immune-mediated adverse events. **Prohibited:** Concurrent investigational agents, chronic systemic corticosteroids (>10 mg daily prednisone equivalent) unless used for managing immune-related adverse events, and other systemic anti-cancer therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -84 to 0	Informed Consent, Surgery Confirmation (within 12 weeks), Scans, Labs, ECOG
Baseline	Day 0	Randomization (1:1), First Dose Administration, Safety Baseline
Interim	Every 3 Weeks	Cycles 2-17: IV Dosing, Safety Labs, AE/irAE Monitoring, Efficacy Scans (per protocol schedule)
End of Study	After Cycle 17	End of Treatment (EOT) Visit, Safety Follow-up, Survival and Recurrence Monitoring (up to 8 years)

(Note: Due to a preplanned futility analysis in May 2024, the study was unblinded and participants on the experimental arm

discontinued the coformulation to be offered pembrolizumab monotherapy).

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Recurrence-Free Survival (RFS), defined as the time from randomization to the first documented disease recurrence or death from any cause. **Measurement Method:** Investigator assessment.

7.2 Secondary & Exploratory Endpoints 1. Distant Metastasis-Free Survival (DMFS) by investigator review. 2. Overall Survival (OS). 3. Safety and tolerability (measured by AE/SAE rates and treatment discontinuations). 4. Quality of Life (QOL).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring throughout the trial. High rates of immune-mediated adverse effects (irAEs) were noted in the combination arm, leading to high discontinuation rates. **Serious Adverse Events (SAEs):** Reported within 24 hours to the sponsor and continuously evaluated. **Data Monitoring Committee (DMC):** An Independent Data Monitoring Committee (IDMC) oversaw safety and efficacy, eventually recommending the unblinding and discontinuation of the combination arm due to the high rate of irAE-driven treatment discontinuations and subsequent RFS futility.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intention-to-Treat (ITT) for primary efficacy (RFS) evaluations. Safety Set for all participants who received at least one dose of the study treatment. **Sample Size Justification:** The trial aimed to randomize ~1,560 patients to sufficiently power a stratified Cox regression analysis for superiority. **Handling of Missing Data:** Standard survival analysis methods including Kaplan-Meier estimates; hazard ratios (HR) and 95% CIs estimated using a stratified Cox regression model with the Efron method of handling ties. An event-driven nonbinding futility analysis was planned at 111 events.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Supervised by an external IDMC to evaluate interim RFS and safety data. **Audit Trail:** Standardized electronic

11. Study Identifiers & Governance

Primary ID: MK-7684A-010 **Registry Number:** NCT05665595 **Sponsor/Lead Organization:** Merck Sharp & Dohme **Study Title:** KEYVIBE-010 **Official Regulatory Title:** A Phase 3, Randomized, Double-blind, Active-Comparator-Controlled Clinical Study of Adjuvant MK-7684A (Vibostolimab With Pembrolizumab) Versus Adjuvant Pembrolizumab in Participants With High-risk Stage II-IV Melanoma (KEYVIBE-010)

12. Clinical Framework (Melanoma Specific)

Disease Area: Melanoma **Preferred UMLS Name:** Malignant Melanoma **Semantic Type:** Neoplastic Process **SNOMED ID:** 372244006 **MeSH Code:** D008545 **HPO Code:** HP:0002861 **Primary Condition:** High-Risk Resected Cutaneous Melanoma (Stage IIB-IV) **Diagnosis Threshold:** Complete surgical resection and histological confirmation **Medical Methodology:** Adjuvant Immunotherapy (anti-PD-1 +/- anti-TIGIT) **Optimization Target:** Recurrence-Free Survival (RFS)

13. Study Procedures & Methodology

Management Pathway: Post-surgical Adjuvant Oncology Pathway **Education Protocol (HCP):** Site initiation visits; training on the management of immune-mediated adverse events (irAEs) **Education Protocol (Patient):** Informed consent processes; monitoring and self-reporting of irAE symptoms **Quality Audit Mechanism:** Independent Data Monitoring Committee (IDMC) interim analyses and safety review

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 8 years (1 year of active treatment followed by extended observation until recurrence, new cancer, or death) **Onsite Visit Cadence:** Every 3 weeks (Q3W) during the 17-cycle treatment phase **Remote Monitoring Frequency:** Intermittent survival and recurrence follow-up checks post-treatment **Checklist Review Interval:** Prior to every Q3W infusion

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				